

A Banach-Lattice Subcase Of Problem 5.8 In arXiv:1506.08013

Automatic math-discovery smoke test

June 29, 2026

Source Problem

The source paper is M. Veraar and L. Weis, arXiv:1506.08013, *Estimates for vector-valued holomorphic functions and Littlewood-Paley-Stein theory*.

The relevant question appears as Problem 5.8 on page 20 of the arXiv PDF. A screenshot crop of the problem is included below.

Let (X_0, X_1) be a compatible interpolation couple and $\theta \in (0, 1)$. For $p \in (1, 2]$ and $q \in [2, \infty)$, on the right hand side of (5.11).

The results of this section lead to the following natural question:

Problem 5.8. Given an interpolation couple (X_0, X_1) and $p \in (1, 2]$ and $q \in [2, \infty)$. Prove or disprove the following:

- (i) If X_0 and X_1 both have type $p \in (1, 2]$, then $(X_0, X_1)_{\theta, p} \hookrightarrow [X_0, X_1]_{\theta}$.
- (ii) If X_0 and X_1 both have cotype $q \in [2, \infty)$, then $[X_0, X_1]_{\theta} \hookrightarrow (X_0, X_1)_{\theta, q}$.

More generally, one can ask for the same result if instead X_j has type p_j and $\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}$ (and similarly in the cotype case).

Problem 5.8 asks whether, for a compatible interpolation couple (X_0, X_1) , endpoint type $p \in (1, 2]$ implies

$$(X_0, X_1)_{\theta, p} \hookrightarrow [X_0, X_1]_{\theta}$$

and endpoint cotype $q \in [2, \infty)$ implies

$$[X_0, X_1]_{\theta} \hookrightarrow (X_0, X_1)_{\theta, q}.$$

The source proves this for couples $(X, D(A))$ when A has a bounded H^∞ -calculus and notes that the corresponding statement is known for the Rademacher interpolation method.

Result Type

This packet is a **partial result**. It proves the expected embeddings for a Banach-lattice subcase, under the natural lattice convexity and concavity hypotheses. It does not answer the full arbitrary Banach-couple question.

Definitions And Setup

Let (X_0, X_1) be a compatible couple of Banach lattices realized in a common vector lattice of measurable functions. Write

$$X_0^{1-\theta} X_1^\theta$$

for the Calderon-Lozanovskii product, with the equivalent norm

$$\|x\|_C = \inf \max\{\|a\|_{X_0}, \|b\|_{X_1}\},$$

where the infimum runs over all positive $a \in X_0$, $b \in X_1$ such that

$$|x| \leq a^{1-\theta} b^\theta.$$

We assume the standard Calderon product identification

$$[X_0, X_1]_\theta \simeq X_0^{1-\theta} X_1^\theta$$

with equivalence constant C_θ . This is the usual situation for standard saturated Banach function lattice couples under the classical Calderon theorem hypotheses.

A Banach lattice X is p -convex if there is a constant $M_p(X)$ such that for every finite family $(u_k) \subset X$,

$$\left\| \left(\sum_k |u_k|^p \right)^{1/p} \right\|_X \leq M_p(X) \left(\sum_k \|u_k\|_X^p \right)^{1/p}.$$

It is q -concave if there is a constant $M^q(X)$ such that

$$\left(\sum_k \|u_k\|_X^q \right)^{1/q} \leq M^q(X) \left\| \left(\sum_k |u_k|^q \right)^{1/q} \right\|_X.$$

We use the discrete Lions-Peetre mean norm for real interpolation. For common exponent r , it is equivalent to the usual K -method norm:

$$\|x\|_{\theta,r} \simeq \inf_{x = \sum_{k \in \mathbb{Z}} x_k} \left[\left(\sum_k e^{-\theta r k} \|x_k\|_{X_0}^r \right)^{1/r} + \left(\sum_k e^{(1-\theta)r k} \|x_k\|_{X_1}^r \right)^{1/r} \right].$$

Idea Of The Proof

The complex method is replaced, in this lattice setting, by the Calderon-Lozanovskii product. The left embedding is then just a weighted Holder inequality: a Lions-Peetre decomposition $x = \sum x_k$ gives

$$|x_k| = (e^{-\theta k} |x_k|)^{1-\theta} (e^{(1-\theta)k} |x_k|)^\theta,$$

and p -convexity controls the lattice p -sums of the two factors.

For the right embedding, start with a Calderon factorization $|x| \leq a^{1-\theta} b^\theta$. Decompose the measure space into level sets where b/a is approximately e^k . On the k -th level set, the piece x_k has the correct weighted X_0 - and X_1 -bounds. Because the pieces are disjoint, q -concavity converts those pointwise level estimates into a real interpolation q -norm bound.

Main Claim

Theorem 1. *Let (X_0, X_1) be a compatible Banach lattice couple for which $[X_0, X_1]_\theta \simeq X_0^{1-\theta} X_1^\theta$. Let $1 < p \leq 2 \leq q < \infty$.*

1. *If X_0 and X_1 are p -convex, then*

$$(X_0, X_1)_{\theta,p} \hookrightarrow [X_0, X_1]_\theta.$$

2. If X_0 and X_1 are q -concave, then

$$[X_0, X_1]_\theta \hookrightarrow (X_0, X_1)_{\theta, q}.$$

The constants depend only on θ , the real-method discretization constants, the Calderon product identification constant, and the relevant lattice convexity or concavity constants of X_0 and X_1 .

Proof. We first prove the p -convex, left-hand embedding. Let $x = \sum_k x_k$ be a finite Lions-Peetre decomposition and set

$$A_k = e^{-\theta k} |x_k|, \quad B_k = e^{(1-\theta)k} |x_k|.$$

Then $|x_k| = A_k^{1-\theta} B_k^\theta$. Hence, by the lattice Holder inequality over the index k ,

$$|x| \leq \sum_k |x_k| \leq \left(\sum_k A_k^p \right)^{(1-\theta)/p} \left(\sum_k B_k^p \right)^{\theta/p}.$$

Put

$$A = \left(\sum_k A_k^p \right)^{1/p}, \quad B = \left(\sum_k B_k^p \right)^{1/p}.$$

The preceding inequality gives a Calderon product factorization $|x| \leq A^{1-\theta} B^\theta$. By p -convexity,

$$\|A\|_{X_0} \leq M_p(X_0) \left(\sum_k e^{-\theta p k} \|x_k\|_{X_0}^p \right)^{1/p}$$

and

$$\|B\|_{X_1} \leq M_p(X_1) \left(\sum_k e^{(1-\theta)p k} \|x_k\|_{X_1}^p \right)^{1/p}.$$

Taking the maximum of the two bounds and then the infimum over all decompositions $x = \sum_k x_k$ proves

$$\|x\|_{X_0^{1-\theta} X_1^\theta} \lesssim \|x\|_{(X_0, X_1)_{\theta, p}}.$$

The Calderon product identification yields $(X_0, X_1)_{\theta, p} \hookrightarrow [X_0, X_1]_\theta$.

We next prove the q -concave, right-hand embedding. Let $|x| \leq a^{1-\theta} b^\theta$ with $a \in X_0^+$ and $b \in X_1^+$. After the usual harmless convention on the sets where $a = 0$ or $b = 0$, partition the underlying measure space into disjoint measurable sets

$$E_k = \{e^k \leq b/a < e^{k+1}\}, \quad k \in \mathbb{Z}.$$

Let $x_k = x \mathbf{1}_{E_k}$. On E_k ,

$$e^{-\theta k} |x_k| \leq e^\theta a \mathbf{1}_{E_k}, \quad e^{(1-\theta)k} |x_k| \leq b \mathbf{1}_{E_k}.$$

Since the E_k 's are disjoint, q -concavity gives

$$\left(\sum_k e^{-\theta q k} \|x_k\|_{X_0}^q \right)^{1/q} \leq e^\theta M^q(X_0) \|a\|_{X_0}$$

and

$$\left(\sum_k e^{(1-\theta)q k} \|x_k\|_{X_1}^q \right)^{1/q} \leq M^q(X_1) \|b\|_{X_1}.$$

Therefore

$$\|x\|_{(X_0, X_1)_{\theta, q}} \lesssim \max\{\|a\|_{X_0}, \|b\|_{X_1}\}.$$

Taking the infimum over Calderon factorizations of x gives

$$\|x\|_{(X_0, X_1)_{\theta, q}} \lesssim \|x\|_{X_0^{1-\theta} X_1^{\theta}}.$$

Using again the Calderon product identification, this is $[X_0, X_1]_{\theta} \hookrightarrow (X_0, X_1)_{\theta, q}$. $\square$

Remark 1. *The same proof gives the mixed-exponent mean-method variant. If X_j is p_j -convex and $1/p = (1-\theta)/p_0 + \theta/p_1$, then the Lions-Peetre space $(X_0, X_1)_{\theta, p_0, p_1}$ embeds into the Calderon product. If X_j is q_j -concave, the Calderon product embeds into $(X_0, X_1)_{\theta, q_0, q_1}$. This matches the more general form of the source question, but still only in the lattice convexity/concavity setting.*

Verification Notes

The proof is analytic and uses no numerical computation. The main points to check are:

- the Calderon product identification for the chosen class of Banach lattice couples;
- the equivalence between the discrete Lions-Peetre mean norm and the standard real K -method norm;
- the direction of the p -convexity and q -concavity inequalities.

All constants are uniform over finite decompositions. Infinite decompositions follow by the usual density/truncation argument in the discrete real method.

Limitations And Novelty Status

This packet does not prove Problem 5.8 as stated. Rademacher type and cotype are isomorphic Banach-space properties, while p -convexity and q -concavity are order-structured lattice hypotheses. The attempted Fourier-block route toward a full counterexample remains unresolved: the missing step is a clean converse showing that a uniform real-to-complex embedding for the Fourier-twisted block couples would force Fourier type.

Cheap run-index searches on June 29, 2026 found the existing no-promotion attempt for arXiv:1506.08013, but no existing solution packet for this scoped lattice result. Local source searches found the adjacent sequential interpolation framework of Lindemulder–Lorist, which records related embeddings between real, complex, gamma, and lattice sequence methods; it does not by itself state this exact Calderon-product partial answer to Problem 5.8.

Human Review Recommendation

This is recommended for human review as a modest but clean partial result. The reviewer should verify the precise lattice hypotheses under which the chosen lower complex interpolation space agrees with the Calderon-Lozanovskii product. Under that standard identification, the proof is short and appears likely valid.

References

- [1] M. Veraar and L. Weis, *Estimates for vector-valued holomorphic functions and Littlewood-Paley-Stein theory*, arXiv:1506.08013.
- [2] A.-P. Calderon, *Intermediate spaces and interpolation, the complex method*, Studia Mathematica 24 (1964), 113–190.
- [3] J. Bergh and J. Lofstrom, *Interpolation Spaces. An Introduction*, Springer, 1976.
- [4] N. Lindemulder and E. Lorist, *A discrete framework for the interpolation of Banach spaces*, arXiv:2105.08373.